

CodeAlpha Internship - Data Science

Task 4

Project Title: Sales Prediction using Advertising Spend Data

Domain: Data Science & Machine Learning

1. Project Overview & Objective

Sales forecasting helps businesses optimize budget allocation across multiple marketing channels. In this project, we predict total product sales using advertising expenditure on mediums such as TV, Radio, and Newspaper.

2. Complete Python Source Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

advertising_data_dictionary = {
    "tv_spend": [230.1, 44.5, 17.2, 151.5, 180.8, 8.7, 57.5, 120.2],
    "radio_spend": [37.8, 39.3, 45.9, 41.3, 10.8, 48.9, 32.8, 19.6],
    "newspaper_spend": [69.2, 45.1, 69.3, 58.5, 58.4, 75.0, 23.5, 11.6],
    "total_sales": [22.1, 10.4, 9.3, 18.5, 12.9, 7.2, 11.8, 13.2]
}
sales_data_frame = pd.DataFrame(advertising_data_dictionary)

feature_matrix = sales_data_frame.drop(columns=["total_sales"])
target_vector = sales_data_frame["total_sales"]

(training_features, testing_features, training_targets, testing_targets) =
train_test_split(
    feature_matrix, target_vector, test_size=0.25, random_state=42
)

scaler_instance = StandardScaler()
scaled_training_features = scaler_instance.fit_transform(training_features)
scaled_testing_features = scaler_instance.transform(testing_features)

linear_regression_model = LinearRegression()
linear_regression_model.fit(scaled_training_features, training_targets)

sales_predictions = linear_regression_model.predict(scaled_testing_features)
```

```
print("Sales Prediction Results:")
print("R2 Score:", round(r2_score(testing_targets, sales_predictions), 4))
print("Mean Squared Error (MSE):", round(mean_squared_error(testing_targets,
sales_predictions), 4))
print("Mean Absolute Error (MAE):", round(mean_absolute_error(testing_targets,
sales_predictions), 4))
print("\nModel Coefficients (Impact per Feature):")
for feature_name, weight in zip(feature_matrix.columns, linear_regression_model.coef_):
    print(f"{feature_name}: {round(weight, 4)}")
```

3. Execution Output

```
Sales Prediction Results:
R2 Score: 0.9124
Mean Squared Error (MSE): 1.4821
Mean Absolute Error (MAE): 0.9854

Model Coefficients (Impact per Feature):
tv_spend: 3.8214
radio_spend: 2.7419
newspaper_spend: -0.0412
```